

List Creation

```
In [1]: list1 = []    # Empty List
```

```
In [2]: print(type(list1))
```

```
<class 'list'>
```

```
In [3]: list2 = [10, 30, 60]    # List of integer numbers
```

```
In [4]: list3 = [10.77, 30.66, 60.89]    # List of float numbers
```

```
In [5]: list4 = ['one', 'two', "three"]    # List of strings
```

```
In [6]: list5 = ['Asif', 25, [50,100], [150,90]]    # Nested Lists
```

```
In [7]: list6 = [100, 'Asif', 17.765]    # List of mixed datatypes
```

```
In [8]: list7 = ['Asif', 25, [50,100], [150,90], {'John', 'David'}]
```

```
In [9]: len(list6)    # Length of List
```

```
Out[9]: 3
```

```
In [10]: len(list7)
```

```
Out[10]: 5
```

List Indexing

```
In [11]: list2[0]    # Retrieve first element of the List
```

```
Out[11]: 10
```

```
In [12]: list4[0]    # Retrieve first element of the List
```

```
Out[12]: 'one'
```

```
In [13]: list4[0][0]    # Nested indexing - Access the first character of the first List el
```

```
Out[13]: 'o'
```

```
In [14]: list4[-1]    # Last item of the List
```

```
Out[14]: 'three'
```

```
In [15]: list5[-1]    # Last item of the List
```

```
Out[15]: [150, 90]
```

List Slicing

```
In [16]: mylist = ['One', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [17]: mylist[0:3]    # Return all items from 0th to 2nd index excludes the 3rd index (n
```

```
Out[17]: ['One', 'two', 'three']
```

```
In [18]: mylist[2:5]    # List all items from 2nd to 4th index Location [(n-1)= 5-1= 4th i
```

```
Out[18]: ['three', 'four', 'five']
```

```
In [19]: mylist[:3]    # Return first three items
```

```
Out[19]: ['One', 'two', 'three']
```

```
In [20]: mylist[:2]    # Return first two items
```

```
Out[20]: ['One', 'two']
```

```
In [21]: mylist[-3:]    # Return Last three items
```

```
Out[21]: ['six', 'seven', 'eight']
```

```
In [22]: mylist[-2:]    # Return Last two items
```

```
Out[22]: ['seven', 'eight']
```

```
In [23]: mylist[-1]    # Return Last item of the List
```

```
Out[23]: 'eight'
```

```
In [24]: mylist[:]    # Return whole List
```

```
Out[24]: ['One', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

Add, Remove & Change Items

```
In [25]: mylist
```

```
Out[25]: ['One', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [26]: mylist.append('nine')    # Add an item to the end of the List  
mylist
```

```
Out[26]: ['One', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']
```

```
In [27]: mylist.insert(9, 'ten')    # Add an item at index Location 9
```

```
In [28]: mylist
```

```
Out[28]: ['One', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine', 'ten']
```

```
In [29]: mylist.insert(1, 'ONE')    # Add an item at index location 1
mylist
```

```
Out[29]: ['One',
          'ONE',
          'two',
          'three',
          'four',
          'five',
          'six',
          'seven',
          'eight',
          'nine',
          'ten']
```

```
In [30]: mylist.remove('ONE')    # Remove item "ONE"
mylist
```

```
Out[30]: ['One', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine', 'ten']
```

```
In [31]: mylist.pop()    # Remove last item of the list by default
mylist
```

```
Out[31]: ['One', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']
```

```
In [32]: mylist.pop(8)    # Remove item at index location 8
```

```
Out[32]: 'nine'
```

```
In [33]: mylist
```

```
Out[33]: ['One', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [34]: del mylist[7]    # Remove item at index location 7
mylist
```

```
Out[34]: ['One', 'two', 'three', 'four', 'five', 'six', 'seven']
```

```
In [35]: # Change value of the string
mylist[0] = 1
mylist[1] = 2
mylist[2] = 3
mylist
```

```
Out[35]: [1, 2, 3, 'four', 'five', 'six', 'seven']
```

```
In [36]: mylist.clear()    # Empty list / Delete all items in the list
mylist
```

```
Out[36]: []
```

```
In [37]: del mylist    # Delete the whole list
mylist
```

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[37], line 2  
      1 del mylist    # Delete the whole list  
----> 2 mylist  
  
NameError: name 'mylist' is not defined
```

Copy List

```
In [79]: mylist = ['One', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']  
  
In [80]: mylist1 = mylist    # Create a new reference "mylist1"  
  
In [81]: id(mylist), id(mylist1)    # The address of both mylist & mylist1 will be the same  
Out[81]: (1936917035264, 1936917035264)  
  
In [82]: mylist2 = mylist.copy()    # Create a copy of the list  
  
In [83]: mylist2  
Out[83]: ['One', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine', 'ten']  
  
In [84]: id(mylist2)    # The address of mylist2 will be different from mylist because copy  
Out[84]: 1936916773120  
  
In [85]: mylist[0] = 1  
  
In [86]: mylist  
Out[86]: [1, 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine', 'ten']  
  
In [87]: mylist1    # mylist1 will be also impacted as it is pointing to the same list  
Out[87]: [1, 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine', 'ten']  
  
In [88]: mylist2    # Copy of the list won't be impacted due to changes made on the original  
Out[88]: ['One', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine', 'ten']
```

Join List

```
In [46]: list1 = ['one', 'two', 'three', 'four']  
         list2 = ['five', 'six', 'seven', 'eight']  
  
In [47]: list3 = list1 + list2    # Join two lists by '+' operator  
  
In [48]: list3  
Out[48]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [49]: list1.extend(list2)    # Append list2 with list1
list1
```

```
Out[49]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

List Membership

```
In [50]: list1
```

```
Out[50]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [51]: 'one' in list1    # Check if one's exist in the list
```

```
Out[51]: True
```

```
In [52]: 'ten' in list1    # Check if ten's exist in the list
```

```
Out[52]: False
```

```
In [53]: if 'three' in list1:    # Check if 'three' exist in the list
          print('Three is present in the list')
        else:
          print('Three is not present in the list')
```

Three is present in the list

```
In [54]: if 'eleven' in list1:    # Check if 'eleven' exist in the list
          print('Eleven is present in the list')
        else:
          print('Eleven is not present in the list')
```

Eleven is not present in the list

Reverse & Sort List

```
In [55]: list1
```

```
Out[55]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [56]: list1.reverse()    # Reverse the list
list1
```

```
Out[56]: ['eight', 'seven', 'six', 'five', 'four', 'three', 'two', 'one']
```

```
In [57]: list1 = list1[::-1]    # Reverse the List
list1
```

```
Out[57]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [58]: mylist3 = [9, 5, 2, 99, 12, 88, 34]
mylist3.sort()    # Sort list in ascending order
mylist3
```

```
Out[58]: [2, 5, 9, 12, 34, 88, 99]
```

```
In [59]: mylist3 = [9, 5, 2, 99, 12, 88, 34]
mylist3.sort(reverse=True)  # Sort list in descending order
mylist3
```

```
Out[59]: [99, 88, 34, 12, 9, 5, 2]
```

```
In [60]: mylist4 = [88, 65, 33, 21, 11, 98]
sorted(mylist4)  # Returns a new sorted list and doesn't change original list
```

```
Out[60]: [11, 21, 33, 65, 88, 98]
```

```
In [61]: mylist4
```

```
Out[61]: [88, 65, 33, 21, 11, 98]
```

Loop through a list

```
In [62]: list1
```

```
Out[62]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [63]: for i in list1:
          print(i)
```

```
one
two
three
four
five
six
seven
eight
```

```
In [64]: for i in enumerate(list1):
          print(i)
```

```
(0, 'one')
(1, 'two')
(2, 'three')
(3, 'four')
(4, 'five')
(5, 'six')
(6, 'seven')
(7, 'eight')
```

Count

```
In [65]: list10 = ['one', 'two', 'three', 'four', 'one', 'one', 'two', 'three']
```

```
In [66]: list10
```

```
Out[66]: ['one', 'two', 'three', 'four', 'one', 'one', 'two', 'three']
```

```
In [67]: list10.count('one')  # Number of item "one" occurred in the list
```

```
Out[67]: 3
```

```
In [68]: list10.count('two') # Occurrence of item "two" in the list
```

```
Out[68]: 2
```

```
In [69]: list10.count('four') # Occurrence of item "four" in the list
```

```
Out[69]: 1
```

All / Any

```
In [70]: L1 = [1, 2, 3, 4, 0]
```

```
In [71]: all(L1) # Will Return false as one value is false (value 0)
```

```
Out[71]: False
```

```
In [72]: any(L1) # Will Return True as we've items in the list with True value
```

```
Out[72]: True
```

```
In [73]: L2 = [1, 2, 3, 4, True, False]
```

```
In [74]: all(L2) # Returns false as one value is false
```

```
Out[74]: False
```

```
In [75]: any(L2) # Will Return True as we've items in the list with True value
```

```
Out[75]: True
```

```
In [76]: L3 = [1, 2, 3, True]
```

```
In [77]: all(L3) # Will return True as all items in the list are True
```

```
Out[77]: True
```

```
In [78]: any(L3)
```

```
Out[78]: True
```

```
In [ ]:
```